AMOGH THAKUR

HARDWARE FOR AI

ECE-510 - CMAN 7

Given Data - values array (1 FP32 4bytes each) column index array (1 INT32 4bytes) row pointer array of length $(N+1) \Rightarrow$ (1 INT32, 4 bytes).

TASK 1 $N = 512$ sparsity s (fraction of zero).

a) Dense MVM (FLOPs).

$\boxed{\text{Expression} - 2N^2 \text{ FLOPs}}$ $\Rightarrow N^2$ multiples. N^2 adds.

For $N = 512 \Rightarrow 2 \times 512^2 = 524{,}288$ FLOPs

b) Dense MVM

$\boxed{\text{Expression} = 4N^2 \text{ bytes.}}$

For $N = 512 \Rightarrow 4 \times 512^2 = 1{,}048{,}576$ bytes $= 1$ MiB.

c) sparse Compute.

$\boxed{\text{Expression} = 2N^2(1-s) \text{ FLOPs}}$

d) sparse Memory. (Addition of value array, column array, row pointer)

Expression $= 2(4\text{bytes} \times N^2(1-s)) + 4 \times (N+1)$ bytes.

$\boxed{= 8N^2(1-s) + 4(N+1) \text{ bytes}}$

TASK 2

$$\text{Speedup} = \frac{2N^2}{[2N^2(1-s)]} = \boxed{\frac{1}{(1-s)}}$$

For speedup 2x s should be 0.5 $s = 0.5$

TASK 3

set spare memory = dense memory.

$$\cancel{8}^{2}N^2(1-s) + \cancel{4}(N+1) = \cancel{4}^{1}N^2$$ ... Divide LHS and RHS by 4.

$$N^2(1-s) = N^2 - (N+1)$$

$$(1-s) = \frac{N^2 - N - 1}{2N^2}$$

$$s = 1 - \frac{N^2 - N - 1}{2N^2} = \frac{2N^2 - N^2 + N + 1}{2N^2} = \boxed{\frac{N^2 + N + 1}{2N^2}}$$

For $N = 512$ $$s = \frac{512^2 + 512 + 1}{2 \times 512^2} = \frac{262144 + 512 + 1}{524288} = \frac{262657}{524288} = \boxed{0.5010 \Rightarrow 50.1\%}$$

TASK 4

$$\text{Speedup at } s = 0.9 = \frac{\text{dense bytes}}{\text{sparse bytes}} = \boxed{\frac{4N^2}{8N^2(1-s) + 4(N+1)}} = \frac{4 \times 512^2}{8 \times 512^2(1-0.9) + 4(512+1)} = \frac{1048576}{211767.2} = \boxed{4.95}$$